PROJECT PROPOSAL

Title:
ParkNow

Introduction
For most people, living in an urban city is often accompanied by the challenge of finding a parking spot. Research indicates that drivers spend an average of 17 hours annually searching for parking spaces [1]. To tackle this issue, our project introduces ParkNow, an innovative web application that will improve the parking experience.

ParkNow uses the power of publicly posted information to provide updates on parking space availability. For example, a user can post the time and location/coordinate whenever an empty parking spot is found to the system, including features of the parking spot like whether it is indoor/outdoor, pricing rates, etc. The web application integrates with Google Maps, offering users a familiar and user-friendly interface for easy navigation. ParkNow has a unique filtering system that enables users to customize their parking search based on specific preferences such as parking conditions and environment, parking fee, or even availability of charging stations for electric vehicles, depending on the information posted by other users. This approach adds a layer of convenience to the parking process, saving users valuable time and reducing the stress of finding parking.

The primary users of ParkNow are urban commuters, people who are routinely driving to crowded cities and urban centers in need of consistent parking spaces. Our design will prioritize usability and user experience. Through intuitive design elements and clear information presentation, ParkNow aims to make the parking experience productive, efficient, and enjoyable. We will be collecting user feedback and conducting usability testing by performing surveys for drivers commuting daily by car in and out of Chicago.

Sketches and diagrams

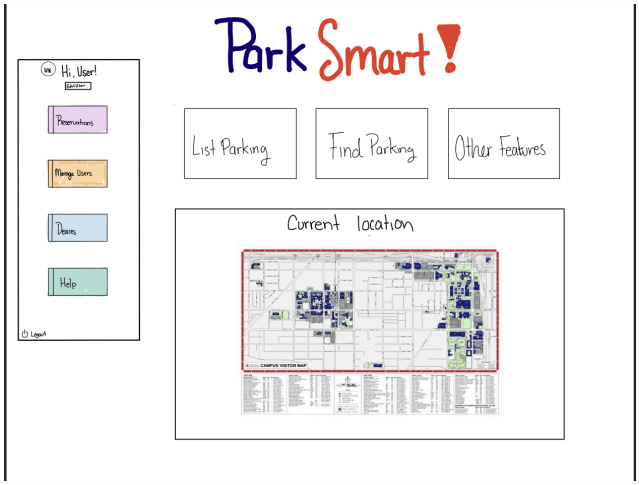

Figure 1. Diagram of the user interface [2]

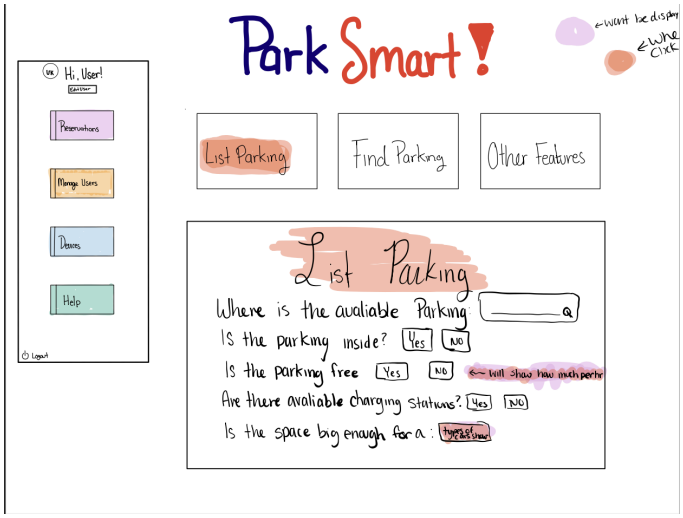

Figure 2. Diagram for List Parking

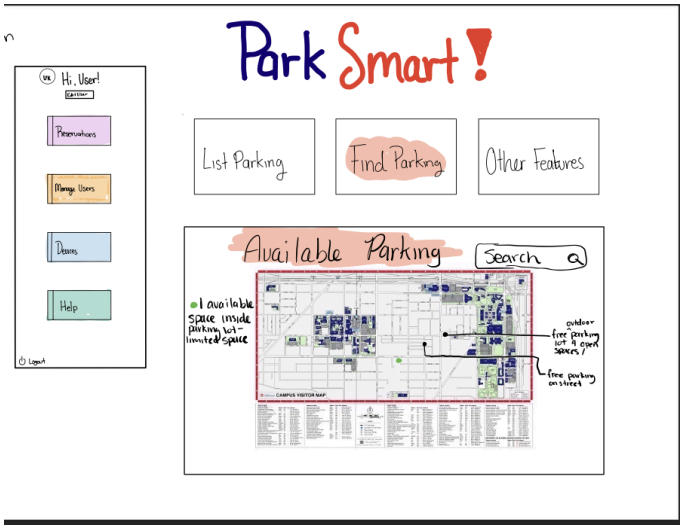

Figure 3. Diagram for Find Parking [2]

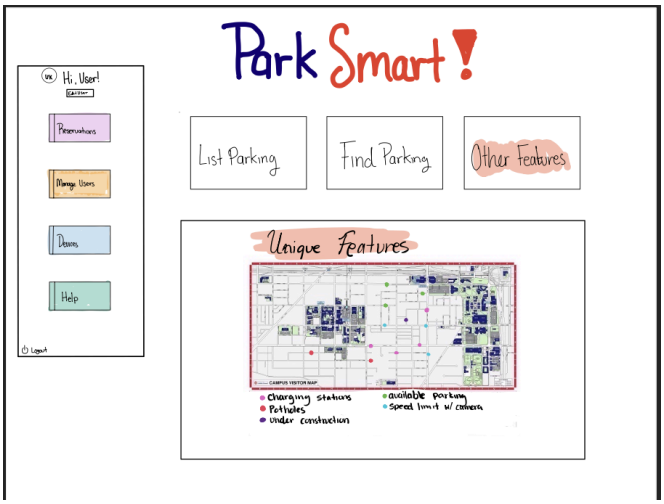

Figure 4. Diagram for Unique Features [2]

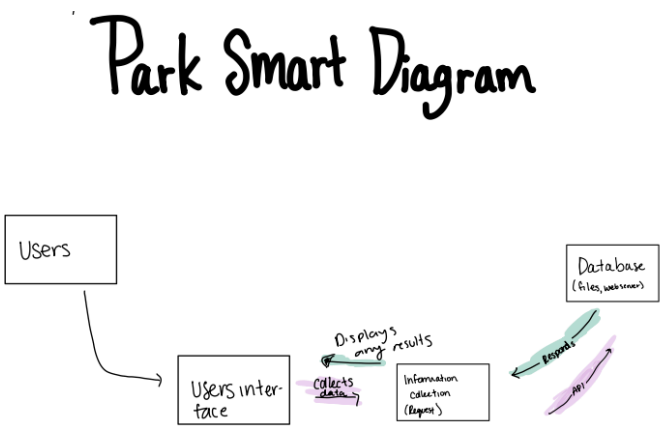

Figure 5. Sequence Diagram for ParkNow

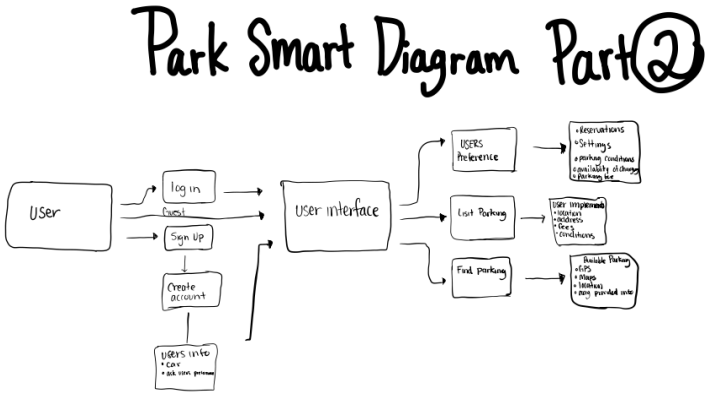

Figure 6. Diagram of ParkNow Website

The name has now changed to ParkNow due to an application with the name “ParkSmart” already existing.

Users and Needs

ParkNow primarily targets urban commuters or occasional city drivers who regularly enter crowded urban areas needing parking. Our surveys will be targeted towards a diverse demographic population, from people living in different parts of Chicago. Finding parking within the city can be difficult and overwhelming, especially for those who don’t have experience within city driving. Users can be found by posting advertisements near commuter areas around UIC, such as the UIC-Halsted Blue Line station, commuter center, community boards and verbally spreading the concept. These locations have many people from different areas of Illinois, which can help spread the application over a broader area. By advertising in commuter hubs around UIC, we aim to broaden our reach.

ParkNow is meant to be a guide for newcomers, but also frequent users. With the creation of electric cars, it has also become important for charging stations to be available for the driver to ensure they can return to their starting location. The application will have the ability to filter out parking locations on whether they have those charging stations or are handicapped accessible. It will be a web-based application that

helps users find open parking spots, but will also have a map embedded within it to help the user find their reserved parking spot. It can be, both, tricky and dangerous for a driver to attempt to switch applications in order to check their parking reservation and use their map application to help find their parking spot. A research study indicated that drivers who use a hand-held device had a slower reaction time and drove constantly slower than driving with no device and with alcohol [3, 4]. With the integration, this could improve safety and convenience. The incorporation of a dataset enriched with latitude and longitude values, through scripts translating addresses, enhances our parking options' accuracy. This integration directly into the map interface, leveraging Google Maps API, uniquely positions ParkNow by improving safety and convenience, offering a seamless parking experience.

Precedents

As technology has advanced, there has been progress made on car devices that can identify and communicate available parking locations. Roughly 10 years ago, a device was patented that would be installed in a car and communicate with parking meter data over the network [5]. As mobile devices were not as advanced as they currently are, having a device installed within the car was the start of mobile parking reservations. Now, technology has progressed to where it is possible for people to reserve almost everything through their smartphone. SpotHero is currently one of the most recent and used mobile parking applications. It’s capable of searching for a spot, selecting a spot, and choosing self park or valet [6]. The app, essentially, gives all possible open options nearby a specified location. SpotAngels is another application that allows similar features for users to find parking spots. Here, it also showcases gas stations nearby.

The intention with ParkNow is to have a more personalized search option, as cars now range in size and capabilities. To differentiate ParkNow from current applications available on the market, we focus on tailoring the parking experience to the individual's specific needs and preferences, such as vehicle size and special features (e.g., electric vehicle charging). Incorporating community-driven updates for parking availability offers real-time insights that may not be available in other apps. Additionally, enhancing the embedded map feature to include detailed navigation within parking lots or structures could offer a distinctive advantage. Offering personalized recommendations based on user behavior and preferences can further enhance the user experience beyond what's currently available. As displayed in Figure 2, a small list of parking preference questions can be given to the user in order to narrow down search options. A map would also be available through the application to avoid the need to switch between applications in order to find how to enter the parking, as shown in Figure 3.

Ingredients

The main components for the ParkNow project will be a combination of software technologies, APIs, and development techniques that will form the foundation for creating our web application. These resources were chosen because they are dependable, scalable, and capable of meeting the unique requirements of our project.

Software Technologies and Frameworks

1. HTML, CSS, and JavaScript: These web technologies will be used to construct the front end of our application. HTML will structure the content, CSS will style it, and JavaScript will add interactivity, ensuring a responsive and engaging user experience.
2. Bootstrap: A front-end framework with an extensive collection of responsive design templates that can speed up the development process.
3. Node.js: Essential for developing the server-side component of ParkNow. It enables us to use JavaScript across the entire development stack, which speeds up development and makes it easier to integrate front-end and back-end components.
4. Firebase: A backend platform that will serve as our database and infrastructure. Firebase provides real-time database functionality, authentication services, and hosting,

APIs

1. Google Maps API: Essential for integrating mapping services into our app. With the help of this API, we will be able to integrate Google Maps, giving users access to real-time location tracking, parking spot visualization, and navigation support.
2. Google Firebase API: Essential for generating and accessing a backend database that would help store user and parking information.
3. Parking Availability API: Essential for finding real-time parking availability to update our database with.

Methodologies

1. Agile Development: To manage the development process, we will adopt an agile methodology, allowing for iterative development, frequent feedback, and flexibility in responding to changes. This approach will ensure that we can quickly adapt to user needs and market trends.
2. User-Centered Design: This is necessary for user experience as it involves engaging with potential users through surveys, usability testing, and feedback sessions to ensure that ParkNow meets their needs and expectations.

Additional Enhancements

1. Data Enrichment Script: A newly integrated script will enhance the dataset with latitude and longitude values, which we get from a reliable parking database API, for precise parking spot identification. An example of a dataset is found at https://mygeodata.cloud/data/download/osm/parking-places/united-states-of-america--illinois/cook-county/chicago.
2. Real-time Reporting Feature: Drawing on similar functionalities in previous applications, this feature will allow users to report on parking spot availability, ensuring up-to-date information.

References:

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- [3] Lipovac, K., Đerić, M., Tešić, M., Andrić, Z., & Marić, B. (2017). Mobile phone use while driving-literary review. Transportation Research Part F: Traffic Psychology and Behaviour, 47, 132–142. doi:10.1016/j.trf.2017.04.015
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- [6] Gebremeskel, G. (2016). *System and method for parking vehicles using a mobile computing device* (U.S. Patent No. 10339749B2). U.S. Patent and Trademark Office. https://patents.google.com/patent/US10339749B2/en?q=(spothero)&oq=spothero